GenePair: Statistical Methods for Modeling Spatially-Referenced Paired Genetic Relatedness Data

SNP Distances Model

$$Y_{ij}|r, p_{ij} \overset{\text{ind}}{\sim} \text{Negative Binomial}\left(r, p_{ij}\right),\ i = 1, ..., n-1,\ j = i+1, ..., n$$
$$\text{logit}\left(p_{ij}\right) = \mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\beta} + \left(\mathbf{d}_i + \mathbf{d}_j\right)^{\mathrm{T}}\boldsymbol{\gamma} + \theta_i + \theta_j$$

$$\begin{aligned}
&\theta_i = \eta\left\{d\left(\mathbf{s}_i\right)\right\} + \zeta_i,\ i = 1, \ldots, n, \\
&\boldsymbol{\eta}^{\mathrm{T}} = \left\{\eta\left(\mathbf{s}_1^*\right), \ldots, \eta\left(\mathbf{s}_m^*\right)\right\}|\phi, \tau^2 \sim \text{MVN}\left\{\mathbf{0}_m, \tau^2\Sigma\left(\phi\right)\right\}, \text{ and} \\
&\Sigma\left(\phi\right)_{ij} = \text{Corr}\left\{\eta\left(\mathbf{s}_i^*\right), \eta\left(\mathbf{s}_j^*\right)\right\} = \exp\left\{-\phi\left\|\mathbf{s}_i^* - \mathbf{s}_j^*\right\|\right\}.
\end{aligned}$$

- $d\left(\mathbf{s}_i\right)$: Maps the spatial location of an individual to an entry within a smaller set of $m < n$ unique locations such that $d\left(\mathbf{s}_i\right) \in \{\mathbf{s}_1^*, \ldots, \mathbf{s}_m^*\}$;
- $\zeta_i|\sigma_\zeta^2 \overset{\text{iid}}{\sim} \text{N}\left(0, \sigma_\zeta^2\right)$;
- m: Number of unique spatial locations ($m \leq n$);
- n: Number of individuals.

Prior Information

$\beta_j, \gamma_k \overset{\text{iid}}{\sim} \text{N}\left(0, \sigma_r^2\right),\ j = 1, ..., p_x,\ k = 1, ..., p_d$;

- p_x: Length of $\mathbf{x}_i$ vector (same for all i) which includes an intercept term;
- p_d: Length of $\mathbf{d}_j$ vector (same for all j) which **does not** include an intercept term;
- Default setting: $\sigma_r^2 = 10,000$.

$r \sim \text{Discrete Uniform}\left[a_r, b_r\right]$;

- Default setting: $a_r = 1$, $b_r = 100$.

$\sigma_\zeta^2 \sim \text{Inverse Gamma}\left(a_{\sigma_\zeta^2}, b_{\sigma_\zeta^2}\right)$;

- Default setting: $a_{\sigma_\zeta^2} = 0.01$, $b_{\sigma_\zeta^2} = 0.01$.

$\tau^2 \sim \text{Inverse Gamma}\left(a_{\tau^2}, b_{\tau^2}\right)$;

- Default setting: $a_{\tau^2} = 0.01$, $b_{\tau^2} = 0.01$.

$\phi \sim \text{Gamma}\left(a_\phi, b_\phi\right)$;

- Default setting: $a_\phi = 1.00$, $b_\phi = 1.00$.

Default Initial Values

- $\beta_j, \gamma_k = 0$ for all j, k;
- $\theta_i = 0$ for all i;
- $\eta_i = 0$ for all i;
- $r = a_r$;
- $\sigma_\zeta^2 = 1.00$;
- $\tau^2 = 1.00$;
- $\phi = 1.00$.

Additional Information

- v: Required input; an n by m matrix with $\mathrm{v}_{ij} = 1$ if individual i resides at location $\mathbf{s}_j^*$ and $\mathrm{v}_{ij} = 0$ otherwise;
- It is recommended to scale the matrix of spatial distances by the largest observed distance in the matrix prior to running the model.